
Suzuki-coupling reaction yield predictor

Ricardo Meireles Rainho, Yusuf Kilicer, Miguel Correia

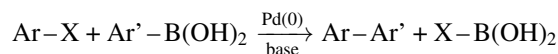
Abstract

Reaction condition optimization in chemical synthesis is still largely based on empirical approaches costly in time and resources. This project offers a ML pipeline capable to predict the yield of a Suzuki-Miyaura coupling and to suggest the best catalytic conditions from reactant structure. The dataset gathers 5760 screened reactions on a HTE platform, covering different combinations of ligands, bases and solvents¹. Reactions are encoded via drfp fingerprints and RDKit molecular descriptors. From this, three models were trained: Random Forest, XGBoost and neural network. The latter reached similar performances, with a R^2 ranging from 0.84 and 0.86. Random Forest obtains the best metric results, but XGBoost was retained for its deployment due to its considerably lower size (~3 Mo vs. ~150 Mo). Two inference tools were developed: an online command script and a Streamlit web app, allowing to identify in a couple of seconds the 5 best chemical conditions among 576 possible combinations for a given pair of reactant.

1 Introduction

Artificial Intelligence applied in chemistry may become fundamental in future research performed in this domain. It already finds uses in many of its different aspects, such as data processing, property prediction, reaction product production, and many other applications. One of the most important uses of AI in chemistry is reaction yield prediction. The yield of a reaction, in percent, is often defined as the experimental yield over the maximum theoretical yield. Having a tool able to predict the experimental yield of a reaction with accuracy is very valuable for chemistry, especially in the field of multi-step organic synthesis. Such a tool allows for the estimation of the total yield across multiple reactions. It can also direct chemists towards reaction pathways that they would not consider otherwise, and encourage them to test reaction conditions never explored before due to empirical speculations or bias.

This project aims to create a script capable of predicting the yield of Suzuki-Miyaura coupling reactions. This reaction involves two reactants, a catalyst, a ligand (X), a base, and a solvent.



Each compound has an effect on the reaction yield. The catalyst used in the dataset is always Pd(0), so it will not be changed. The final pipeline should take as input two SMILES corresponding to the reactants and return a list of the five best reaction conditions according to their predicted yield.

For that, the data is treated, featurized, and converted to optimized training and testing sets. Models are then fitted to these and used in the final script to complete the task.

2 Task

The task is to produce a regression model that, given two reactants represented as SMILES and included in the dataset, predicts the yield of the Suzuki-Miyaura reaction between them. The model

tests all possible combinations of ligands, bases, and solvents present in the dataset. For a given combination, the model tests 576 possible catalytic system combinations and returns the 5 reactions with the best predicted yield. Therefore, the model does not attempt to optimize the combination itself, but rather the yield; it evaluates the reactions individually, and the selection is made indirectly based on their yield.

3 Methods

3.1 Data treatment

The dataset used for this project comes from the study by Perera et al. (Science, 2018)¹, which compiled the results of a high-throughput synthesis platform for Suzuki-Miyaura coupling reactions. It contains 5856 reactions that differ in the reactants, ligand, base, and solvent. The catalyst used is constant; it is the catalytic system that changes. The performance of the catalytic system is evaluated according to the yield measured by HPLC-UV.

However, this data required processing. First, the missing values in the base and ligand columns simply represented reactions in which the base or ligand was absent. After analysis, 480 reactions were performed without a ligand and 720 without a base. In these reactions, the ligand or base was renamed "None" to indicate its absence, and its molar equivalent was set to 0, allowing the model to learn the effect of the absence of these components.

Next, columns with no variance were removed. Indeed, they would not contribute anything to the model; they constituted superfluous information. Among these, the column for the catalyst was removed, as its value (Pd(OAc)₂) remained constant throughout all experiments. The equivalents of the reactants and the catalyst also remained constant and were therefore also removed. Finally, a column representing the yield measured by mass spectrometry was also removed because it was deemed unreliable.

After analyzing the different solvents used, the presence of THF.V2 was observed in 96 experiments. Chemically identical to THF, its reactions showed a median yield difference of 12%. It was therefore decided to remove the reactions containing this solvent, which represented only 1.7% of the dataset, thus reducing noise.

Finally, the distribution of the yield measured by HPLC-UV was verified. No values outside the range [0;100]% were detected. However, a large proportion of reactions had zero yield. The final dataset contains 5760 reactions.

3.2 Featurization

The feature matrix is constructed by combining two complementary representations for each Suzuki coupling. The reagents (electrophilic and boronate nucleophilic partner) are encoded with the *Differential Reaction Fingerprint*², which calculates the symmetrical difference of Morgan-type circular footprints (maximum radius 3) between the reactants and the products, folded into a binary vector of 2048 bits. This representation explicitly captures the formed and broken bonds, which are information directly correlated to the coupling's success.

The ligand, base and solvent, absent from the reaction SMILES, are encoded separately with the RD-Kit molecular descriptors. For multi-fragment SMILES (salts, complexes), only the largest organic fragment is retained before calculation.

The four blocks (DRFP + ligand + base + solvent) are concatenated into a crude matrix of dimension (n , 2675). The column containing NaN, infinite or zero variance values are deleted. The yield is normalized within the [0, 1] range. The dataset is then divided into train/test (80/20), and a `StandardScaler`, exclusively tailored on the whole of the training, is applied to center and reduces each feature while avoiding any leakage of information towards the test.

3.3 Models

For yield prediction, two regressors and a neural network model are used.

The regressors are RandomForest and XGBoost. Both were trained using GridSearch to try and find the best combination of hyperparameters.

The RandomForest regressor is a model using multiple trees that are each trained on a different subset of the dataset. After being fitted to the training data, the model takes as input a vector and returns an average of each value given by each tree. The XGBoost regressor is constructed by building multiple decision trees successively. Each tree tries to adjust and correct the imprecision of the one before. It uses gradient based optimization to improve its performance.

The neural network model makes use of multiple interconnected node layers. Each one applies a function to its inputs with different weights and sends the output to the next layer. The last node takes the outputs of all the previous layer's nodes and calculates a single value, which is the predicted reaction yield.

The neural network has 3 hidden layers with a dropout rate of 0.4, a hidden size of 128 and a learning rate of $1e-4$, since the model training is relatively quick.

The models are trained on the optimized data and on random splits with different training fractions of the complete set. This is done to identify whether this method of training is more effective than the former.

4 Results

The model performance was assessed through the calculation of performance metrics such as the MAE, the RMSE, the R^2 score, and the explained variance score. Many plots were drawn in the code to allow for better visualization and to identify potential sources of error. Here is a table of the metrics along with the parity plots for each trained model.

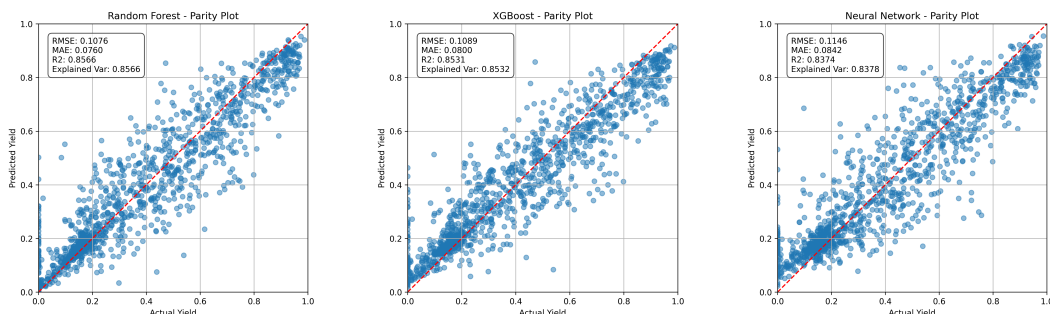


Figure 1: Parity plots of RandomForest, XGBoost, and Neural Network respectively.

Model	RMSE	MAE	R^2	EV
RandomForest	0.1076	0.0760	0.8566	0.8566
XGBoost	0.1089	0.0800	0.8531	0.8532
Neural Network	0.1146	0.0842	0.8374	0.8378

Table 1: Comparative metrics table for the three models. The metrics are Root Mean Squared Error, Mean Absolute Error, R^2 score, and Explained Variance score respectively.

All of the models have similar parity plots. The neural network model shows a slightly sparser plot, which is correlated with a lower precision. There is no inconsistency in the plot other than the fact that all the models seem to sometimes overestimate yields when the actual one is near zero.

The regressor models have similar metrics, with the RandomForest having slightly better ones overall. The Neural Network has worse metrics overall, but the difference is not dramatic.

The models have trouble identifying low yield reactions, but looking at the plots, the models don't overestimate them too much. This tendency should not cause a problem if the models are mainly used to find high yield reactions.

The models tend, however, to underestimate the yield a bit when confronted with high yielding reaction conditions. It is shown in the GitHub repository that their precision when outputting high yields is relatively high. This may indicate that the models, when predicting a high yield, are likely to be correct, which is a desirable property.

In terms of metrics and performance, the RandomForest regressor is the best model, closely followed by XGBoost. The preferred model would be XGBoost, nevertheless, because it has a very similar performance to RandomForest while occupying a lot less space, making it likely to perform quicker predictions. In fact, the RandomForest regressor takes around 150 Mo while the XGBoost regressor takes less than 3 Mo.

5 Conclusion

The objective of this project is to automate the selection of the optimal catalytic system for Suzuki-Miyaura ligand-base-solvent coupling, using two reagents in SMILES form. The different models tested were trained on a dataset from a high-throughput experimental platform.

The three trained models were RandomForest, XGBoost, and a neural network. These models exhibited comparable performance, with an R^2 between 0.84 and 0.86 across the entire test. The combined DRFP representation and RDKit molecular descriptors proved effective in encoding both the chemical transformation and the physicochemical properties of the reaction conditions. Although RandomForest displayed the best metrics, XGBoost was selected for deployment due to its significantly smaller disk footprint, nearly fifty times smaller, while offering very similar performance.

However, this project is limited by several aspects that deserve highlighting. The models underestimate very high yields and overestimate yields close to zero. This behavior is expected when applying regressions to data with a skewed distribution. Furthermore, the dataset is restricted to 28 reactant pairs from the HTE with a constant catalyst. The model can hardly be generalized to new substrates or other reaction families.

Integrating new substrates and data from the literature would improve the robustness of predictions. It is also possible to integrate graph neural networks to directly encode molecular structures, which is a promising avenue.

In conclusion, this project demonstrates that machine learning can guide the optimization of response conditions, offering a fast and inexpensive alternative for research.

References

- [1] Damith Perera, Joseph W. Tucker, Shalini Brahmhatt, Christopher J. Helal, Ashley Chong, William Farrell, Paul Richardson, and Neal W. Sach. A platform for automated nanomole-scale reaction screening and micromole-scale synthesis in flow. *Science*, 359:429 – 434, 2018. URL <https://api.semanticscholar.org/CorpusID:5003014>.
- [2] Daniel Probst. Commit: Reaction classification and yield prediction using the differential reaction fingerprint drfp. *Digital Discovery*, 4(8):1988–1990, 2025. ISSN 2635-098X. doi: <https://doi.org/10.1039/d5dd00089k>. URL <https://www.sciencedirect.com/science/article/pii/S2635098X25001196>.
- [3] Philippe Schwaller, Alain C. Vaucher, Teodoro Laino, and Jean-Louis Reymond. Prediction of Chemical Reaction Yields using Deep Learning. 2020(1012). doi: 10.26434/chemrxiv.12758474.v2. URL <https://chemrxiv.org/doi/full/10.26434/chemrxiv.12758474.v2>.